

Troubleshooting

Every entry below is symptom -> check -> fix, grounded in the code that ships in this repo, not in how a hook is supposed to behave. Where a command is shown as `<crew>/hooks/scripts/...`, that is `${CLAUDE_PLUGIN_ROOT}/hooks/scripts/...` inside a command file, or the plugin's cache directory when you run it by hand — see the shorthand in [Memory and Obsidian](#).

Stale Install

Symptom: a command from this guide (`/crew:status`), a new hook, a config key) does not exist, or behaves like an older release.

- **Check the installed version against the repo.**

```
cat ~/.claude/plugins/installed_plugins.json | grep -A3 '"crew"'
```

Compare the commit or version it records against `plugin/crew/.claude-plugin/plugin.json`'s version in the marketplace clone. `claude plugin update` decides whether to re-copy a plugin by comparing the **declared version**, not its content — a content change that ships with no version bump means every already-installed copy reports "already at the latest version" forever, holding the old code, with nothing in the repo itself looking wrong (`CLAUDE.md`, "Stop and ask"). This class of bug is why `/crew:status`'s absence, or a hook that clearly predates a documented fix, is worth checking here before assuming the fix is wrong.

A related, sharper case: on a directory-source install (a dev checkout registered as a local plugin path rather than pulled from a marketplace), `installed_plugins.json` can name a version string with **no relationship to the code that actually executes** — the tree runs whatever is on disk regardless of what that file says (`TOD0.md`, "`installed_plugins.json` records 0.19.56 while 0.19.63 executes"). On that install shape, trust the code over the version string.

- **Fix:**

```
claude plugin update crew@useful-claude-add-ons
```

then **restart** `claude` — a running session does not pick up a re-copied plugin. If the update reports "already at the latest version" and you can see the fix in the marketplace repo's `main`, the version was not bumped when it should have been; that is a defect to file against the plugin, not something a client-side retry fixes.

- **Check it worked:**

```
/crew:status
```

Read-only, at most 40 lines, dispatches nothing. Its config line names which config file it read (.crew/crew.json for 1.0, .crew/config.json for 0.20); codemap reports behind: <subsystem> when a map's anchor and the path diff both say it is stale.

Hook Noise

Every hook crew registers, from plugin/crew/hooks/hooks.json, and the config key that turns its behaviour up or down. All events are registered once for bash and once for PowerShell (shell: "powershell"), so a Windows machine with both bash and pwsh on PATH fires both halves of every pair — that is what "even counts" in the file means, not a bug.

Event	Script	What it does	Switch	Default
SessionStart	handoff-read.sh	resets its once-per-session markers; reads back .work/HANDOFF.md only when memory.inject is false	— (always runs)	on
SessionStart	platform-sync.sh	detects/repairs platform. {os,wsl,shell,windowsHostIp}	— (always runs)	on
SessionStart / UserPromptSubmit / PostToolUse / SubagentStart	crew-context.sh	injects code-map slices and vault recall, tracks the token budget	context.enabled (whole hook), memory.inject (codemap, handoff and vault injection — on by default since 1.0.0)	context.enabled true, memory.inject true
UserPromptSubmit	approval-hook.sh	records or refuses a /crew:approve <id> receipt	always records; enforcement depends on scope.mode	n/a
PreToolUse	promote-gate.sh	the six	guards.terraformApply,	block / no

Event	Script	What it does	Switch	Default
(Bash/PowerShell)		command/production guards plus mergeGate	guards.forcePush, guards.adminMerge, guards.mergeGate, guards.cloudDestructive, guards.sqlDestructive, guards.prodDatabase, guards.prodServer	strictest
PreToolUse (Write Edit)	role-write-guard.sh	refuses a write outside the dispatched role's declared scope	guards.roleWrites (block/report/off)	off
PreToolUse (Bash PowerShell)	cloud-guard.sh	destructive aws/az commands and wrong-identity commands	guards.cloudGuard (block/report/off)	off
PreToolUse (Write Edit MultiEdit NotebookEdit Bash PowerShell)	scope-guard.sh	plan-approval + ticket scope guard	scope.mode (off/report/block/auto), scope.allowCliApproval	off, false
PreCompact	handoff-write.sh	writes the handoff note before compaction	context.autoWrapUp, context.handoffPath	on
Notification	notify.sh	pings an external channel	notify.provider (none disables it), notify.events	none
Stop	verify-gate.sh	runs .crew/verify.json's checks	verifyGate (boolean)	true
Stop	context-watch.sh	nags for a handoff near the context	context.enabled, context.warnAt, context.budgetTokens,	on, warnAt

Event	Script	What it does	Switch	Default
		budget, drives auto-clear	context.reserveTokens, context.autoClear.*	
Stop	completion-audit.sh	diffs the whole tree against the ticket's scope base	scope.mode	off

- **Silence one hook without touching the rest:** set its own key. `guards.roleWrites: off`, `guards.cloudGuard: off` and `scope.mode: off` are already the shipped defaults — a noisy session usually means one of these was turned on somewhere (repo or machine-global) and forgotten, not that the default changed.
- **Silence the SessionStart context specifically:** `{"memory": {"inject": false}}` stops the context hook; `handoff-read.sh` then prints the handoff on resume instead.
- **See what the context hook is actually doing:**

```
python3 "<crew plugin root>/hooks/scripts/crew_context.py" --stats
```

Reports emissions, injected characters, dedup hits, and vault-recall hit/miss/skip counts with a reason breakdown (see "Obsidian" below). The log itself is `<git-common-dir>/crew/context-log.jsonl` (or `.work/crew/context-log.jsonl` outside git), shared across every worktree of the repo. - **A hook fails and you cannot tell why:** every wrapper here resolves its own Python (`python3/python/py`) rather than assuming one is present, and every one that can block fails **closed** — never silently — when no interpreter resolves. See "no-python fail-closed" under Scope, below; the same shape applies to `scope-guard.sh`, `role-write-guard.sh`, `cloud-guard.sh` and `verify-gate.sh`.

Review Loops

Each ticket gets **two** review rounds, reserved before the reviewer runs, tracked in `<git-common-dir>/crew/review/<ticket>.json` — shared by every worktree of the repo, so a second worktree of the same repo spends the same budget (`review_ledger.py`).

- **Symptom: review keeps re-running / "budget exhausted". Check:**

```
python3 "<crew>/hooks/scripts/review_ledger.py" --root . --ticket <id> --status
```

A round's outcome is `CLEAN`, `FINDINGS` or `INCOMPLETE`. A completed round 2 — either `FINDINGS` or `INCOMPLETE` — leaves the ticket state `REVIEWED`, so its `FINDINGS` can still be

accepted. A **third** reservation attempt is refused outright and the state becomes NEEDS_REPLAN; that refusal, and an explicit --reject, are the only two ways into NEEDS_REPLAN. **Fix:** a final round with 0 BLOCK from a Codex or Kimi reviewer closes itself: review: auto-accept: eligible, then --auto-accept --follow-up <id> writes an auto-accepted receipt and its FIX/NIT lines go verbatim into one follow-up ticket. A review: auto-accept: refused - <reason> line names what stopped it (any BLOCK, INCOMPLETE, not the final round, an open healer skip, a count it could not read, a same-family Claude-fallback round, a provider or family it could not tell, or a verdict recovered from stray lines - ignored_lines above 0, or missing or unreadable, which is could-not-tell, as on a round recorded before L-0576 wrote the count). Otherwise own the FINDINGS with --accept --by <who> (only the most recent completed round, only once, never once NEEDS_REPLAN; a name starting auto: is refused), or write a new plan and get it approved — crew_ticket.py approve on a NEEDS_REPLAN ticket opens a fresh budget of two rounds counted from the successor plan; the rounds already spent stay in the ledger and are not erased.

- **Symptom: a round came back INCOMPLETE. Check:** the review: lines, or failure_class in .work/tickets/<id>/review.json. An INCOMPLETE round has one of three classes. tool means the answer never arrived intact: a timeout, a bad or unknown exit, empty output, or a failed Codex stream. tree means a bundle part or web-test report changed under the reviewer. reviewer means the output arrived and broke the contract. Harmless prose, a heading or a code fence beside well-formed findings, with every part acknowledged and exit 0, no longer does that: the round is FINDINGS, and the ignored lines are printed on a review: FINDINGS kept; ... line and kept in review.json's ignored_text (ignored_lines is their count). A stray line beside CLEAN, a misformatted contract line (- FIX|..., fix|..., a | table row) or a line admitting the review fell short ("skipped", "truncated", "could not review") is still INCOMPLETE reviewer. **Fix:** a tool round is refunded automatically, up to two per plan. The line reads review: round N was a tool failure (...); refunded. Only the failed round is given back: the rerun /crew:review reserves a new round, charged like any other unless it is a tool failure too, so a ticket with one charged round that reruns and gets FINDINGS has spent the budget. If Codex is out of quota, use the next eligible provider. A third tool failure under one plan reads NOT refunded - refund limit 2 per plan reached and counts. A reviewer or tree round always counts. Rebuild the bundle (tree), or rerun and read what the reviewer wrote (reviewer).
- **Symptom: /crew:done refuses with a receipt error. Check:**

```
python3 "<crew>/hooks/scripts/review_ledger.py" --root . --ticket <id> --check-receipt
```

A CLEAN verdict writes a receipt automatically; a FINDINGS verdict only becomes one

through `--accept`, or `--auto-accept` on a final 0-BLOCK round. An auto-accepted receipt stands only while its round still reads 0 BLOCK with the same lines from the same Codex or Kimi reviewer the receipt names. `--check-follow-up` fails until the follow-up's `direction.md` quotes every line verbatim (a line the receipt carries twice, twice), and on a non-UTF-8 file or an unknown receipt kind. `/crew:done` does not run it yet (L-0568 adds it to check 1), so run it yourself before closing. `--check-receipt` rebuilds the review bundle from the receipt's recorded base and fails unless the hash still matches, the receipt is for the **latest** recorded round, and the state is not `NEEDS_REPLAN` — so editing a file after the reviewer read it, or after the receipt was written, invalidates the receipt even though nothing about the ledger itself looks wrong. **Fix:** if the edit was deliberate, get the ticket reviewed again (spends the next round); if it was accidental, revert the edit and re-check.

- **Symptom: Codex hit a usage limit.** The probe printed `PROBE=limited` (exit 5) with the error on `PROBE_DETAIL=...`, or a round printed `review: codex usage limit in round N: ....` **Cause:** Codex's usage limit, rate limit, quota or spend cap - its own message is quoted. **What happens:** that round runs on the Claude reviewer, announced as same-family (codex limit) and not independent; the limit is recorded in `<git-common-dir>/crew/review-limit/<ticket>.json` and applies to the next round only. The round is spent like any other. To check Codex yourself without spending one:

```
python3 "<crew>/hooks/scripts/review_run.py" --root . --ticket <id> --scratch  
<dir> \  
  
--provider codex --probe
```

- **Symptom: you want to send a ticket back without spending the third round.**

```
python3 "<crew>/hooks/scripts/review_ledger.py" --root . --ticket <id> --reject --by  
<who>
```

Refuses on a ticket already `ACCEPTED` or already `NEEDS_REPLAN`.

- **Symptom: `crew_train.py` acquire exits 1, waiting behind <ticket>.** The clone's merge train is armed (L-0520) and an overlapping ticket holds it, or queued first on the same base. **Check:**

```
python3 "<crew>/hooks/scripts/crew_train.py" --root . status
```

Each waiting entry lists the ticket it is behind and every colliding pair (`<mine> x <theirs>`); touch: undeclared: `<why>` means that ticket's spec has no usable `## Touch`, which overlaps everything. **Fix:** wait for the holder to land and release, then acquire again before

reviewing; fix an undeclared Touch in the spec. merge <base> first means the base moved in this ticket's Touch: run `crew_train.py catch-up --ticket <id>` and review the merged head. `could not tell (exit 3)` means the train state could not be read — the message names the file; nothing is guessed.

- **Symptom: a lane holds the train and its session died.** `status` prints `stale?:` beside it (worktree missing, head already in the base, held for hours). Nothing releases it automatically, by design. **Fix:** once you are sure it is dead:

```
python3 "<crew>/hooks/scripts/crew_train.py" --root . release --ticket <id> --
force \

--by <who> --reason "<why>"
```

The release is logged as a force-release event that the other lanes see.

- **Symptom: check-land says the base moved in Touch paths.** Another ticket landed changes to paths this ticket touches after it was gated, so the verdict covers a different tree. **Fix:** `crew_train.py catch-up --ticket <id>` (a merge; conflicts and rerere-replayed files are listed and left unstaged for you to inspect, `git add` and `commit`; a version file is never replayed and comes back conflicted), gate the merged head again (`/crew:review`), then `check-land` again. `merge-tree: HEAD conflicts with <base>` is the same fix with a conflict to resolve first.

Scope: Approval and the Completion Audit

A ticket's contract lives in `.work/tickets/<id>/: spec.md`'s `## Touch` section names every path the ticket may change; `plan.md`'s `Files:` lines must each fall inside Touch. Two hooks hold a session to that contract — see [Daily workflow: scope and approval](#) for the contract itself. This section is what goes wrong with the approval and the audit.

- **Symptom: a lane worktree does not see my settings** (a CLI approval refused, `scope.mode` read as `off`, guards at their defaults). `.crew/*` is gitignored, so `git worktree add` makes a checkout with no crew config. **Check:** `/crew:status` prints `config inherited from the main checkout (<path>)` ... when the worktree reads the main checkout's `.crew/config.json` (crew 1.0.69+, T-0088), and `crew_config.py --root . --explain` starts with `repo layer: <path> (<source>)`. **Cause and fix:** a worktree with its own `.crew/config.json` or `.crew/crew.json` reads only those, never merged with the main checkout's - delete them to inherit. `/crew:status` names that case: ... the main checkout's `(<path>)` is not read A lane made before 1.0.69 almost always has one, a default that crew's `SessionStart` heal wrote there. `(unknown)` means git could not name the main checkout; then no default is written

either. The shell and PowerShell readers (verify-gate.sh, _common.sh, notify.sh, the handoff scripts, promote-gate.ps1, scope-guard.ps1, cloud-guard.ps1, auto-clear.ps1) do not inherit yet.

- **Symptom: an edit inside Touch is still refused. Check:** approval status.

```
python3 "<crew>/hooks/scripts/crew_ticket.py" status --ticket <id>
```

Returns approved (both spec.md and plan.md hash to what was approved), stale (either file changed since), or none. **Editing spec.md or plan.md after approval makes the approval stale immediately** — that is deliberate: an edited spec cannot silently widen what the guard accepts. Only a receipt from the user's own prompt counts by default; a cli-sourced receipt (written by crew_ticket.py approve directly, for tests/CI) is accepted only when .crew/config.json sets scope.allowCliApproval: true, and an autopilot receipt (from crew_autopilot.py approve) only while that is true and autopilot.approval still allows it. **Fix:** re-approve. The user types /crew:approve <id> again — the only other route is /crew:autopilot under an opted-in autopilot.approval; scope_guard.py refuses a Write/Edit under <git-common-dir>/crew/ in every mode but off, so a session cannot forge or refresh its own approval.

- **Symptom: you need to touch one more path mid-ticket. Fix:** amend spec.md's ## Touch (widen it), then approve again. There is no partial-approve; amending scope is edit-then-approve, same as any other spec change.
- **Symptom: /crew:done refuses on "out of scope" for a file the edit guard never saw.** This is the Stop-time completion audit (completion_audit.py), not the edit guard. The edit guard only sees Write/Edit/MultiEdit/NotebookEdit tool calls; a sed -i, a redirect, a formatter or a git mv never reaches it. The audit instead diffs the **whole working tree** against the ticket's scope base (scope_base.resolve — the commit the ticket started from) across committed, staged, unstaged and untracked changes, so a shell-made write is caught here even though nothing blocked it at the time. A file byte-identical to main as last merged is not counted; a merged main: could not tell line (a detached HEAD, none of origin/HEAD, origin/main and main naming a commit, or a git error) means every merged-in file was counted, so check out the ticket branch and rerun. A missing origin/main alone is not that: a local main is used instead. An untracked merged-in file (after git rm --cached) is main's only when git add would record it identically: with core.fileMode=false its execute bit is ignored, as git ignores it. On the review side, diffed-from-merged=could-not-tell (or fork: could not tell from --check-receipt) means git merge-base <start> <merged> gave no answer, so a file main also changed shows main's lines as the ticket's: fetch, check the start commit still exists, and rebuild. **Check it directly, without waiting for a Stop:**

```
python3 "<crew>/hooks/scripts/completion_audit.py" --check --ticket <id>
```

Fix: either the path genuinely needs to be in scope (widen Touch and re-approve), or revert the out-of-scope change. Files git ignores (including everything under `.crew/`) and `.work/` itself are outside what the audit can see at all — that is by design, not a gap to work around.

- **Symptom: the scope base is the merge-base with main on a repo whose branches come from development.** `scope_base.py --record` says "(fallback) recorded ... the merge-base with origin/main", and the review bundle or the completion audit lists hundreds of files the ticket never touched. With no `tickets.baseBranch`, the base branch is `origin/HEAD`'s target, then `origin/main`, then `main`, so a branch cut from development is measured against `main` and the whole integration branch looks like this ticket's change. **Fix:** set `"tickets": {"baseBranch": "development"}` in the repo's `.crew/config.json`, then `scope_base.py --root . --record <id>`. A fallback recorded against the old branch is re-derived ("re-derived ... was a merge-base guess against origin/main"); an exact record is never moved. A value that names no commit here, or a config that does not parse, reads as **could not tell**: `--record` exits 1, `--base` exits 3 with nothing on stdout, and the audit fails. It never falls back to `origin/HEAD` silently. Fix the value; do not unset it to make the error go away.
- **scope.mode values, and what "auto" means:** `off` (hooks do nothing, the default), `report` (allows everything, logs the row to `.crew/guard.log`), `block` (refuses out-of-scope writes and fails the completion audit), `auto` (report for the first ten tickets approved in this repo, then `block`). A config file that exists but fails to parse, or names an unknown mode, resolves to `block` and says so — the unknown never collapses into the permissive value.
- **No-python fail-closed, and why it is not a bug:** if none of `python3/python/py` resolves to a *proved* working interpreter, `scope-guard.sh` (and `completion-audit.sh`, byte-for-byte the same logic) does **not** silently allow the write. It checks whether `.crew/config.json` **provably** sets `scope.mode` to `off` — a crude, non-JSON textual check, because it runs in exactly the situation where python cannot parse the file for it — and only then exits 0. Anything it cannot prove `off` (a corrupt file, `report`, `auto`, `block`, or no `scope` key at all) fails **closed** with exit 2: "no usable python ... failing closed". Fix by installing a real Python 3, not by reading the closed refusal as a false positive.

Promote Gate Blocks a Worktree Deploy

`promote-gate.sh / .ps1`, the `PreToolUse` hook on declared `deploy` commands. Since T-0505 it

judges **the tree the deploy runs from**: the Bash call's `cwd`, moved by a leading `cd <dir> &&` and named by any `git -C <dir>`. That tree must be a worktree of the same repository, clean, and at every literal sha the command names. `.crew/verify.json`, `.work/PROMOTIONS.md` and the `.crew/.approved-<env>-<sha>` markers are still read from the session's project directory.

- **"the tree this deploy runs from ('...') is dirty"** names the tree it judged. If that is the main checkout while you meant a worktree, run the command there: `cd <worktree> && <deploy>`, or enter the worktree first. A dirty main checkout no longer blocks a clean worktree.
- **"no all-pass row for sha X"** where X is the worktree's sha: the upstream environment passed a different sha. Promote the worktree's sha upstream first; a row for the main checkout's sha does not carry over.
- **"the command names commit '...', but the tree it runs from ... is at ..."**: a literal `ref=<sha>` that is not the tree's HEAD. Deploy from a tree at that sha, or drop the literal.
- **"changes directory after it starts"** (a later `cd`, `bash -c 'cd ...'`, `env -C`, `make -C`), **"names more than one tree"**, **"not a form the gate reads with certainty"** or `--git-dir`: the gate will not guess. Put a single literal `cd <dir> &&` first, or use `git -C <dir> -` and note the directory the command itself runs in must be clean too.
- **"index entries flagged skip-worktree or assume-unchanged"**: those flags hide edits from `git status`; clear them in the tree being deployed.
- **"has uncommitted changes: it ..."** also fires when the map is deleted or untracked, or edited under `skip-worktree`: the gate compares the file with HEAD's copy, not with `git status`.
- **".crew/verify.json in the project dir ... has uncommitted changes"**: commit or revert the map; an uncommitted map is not policy.
- Never route around a block by running the deploy yourself with `!. /crew:promote` fixes the precondition the message names and asks you only for a `requireHuman` yes or a genuinely interactive step.

Cloud Guard False Positives

`cloud-guard.sh` / `cloud_guard.py`, a `PreToolUse` hook on Bash and PowerShell. **Ships off** (`guards.cloudGuard: off`); turn it on with `{"guards": {"cloudGuard": "report"}}` first to see what it *would* refuse before enforcing with `"block"`.

- **SQL text that merely looks destructive is not blocked.** `DROP/TRUNCATE` inside a string literal or a comment does not count: the guard strips literals and comments before matching, read both as standard SQL and as MySQL, because the two

dialects disagree about what `\` and `--` mean — a payload is flagged only if **either** reading still shows the keyword outside a literal. So `psql -c "SELECT 'DROP TABLE x' AS s"` and `psql -c 'SELECT 1 -- DROP TABLE x'` are both allowed (plugin/crew/tests/test_cloud_guard.py, cases sql-literal/sql-comment/sql-block-comment). If a query like this is still blocked, the reason (`[sqlDestructive]` in the deny message) is worth reading closely — the keyword may genuinely sit outside every literal reading.

- **A dry-run, help, or -WhatIf invocation is still classified as destructive.** The guard matches on the *subcommand*, not on flags that make it a no-op: `terraform destroy --help`, `aws ec2 terminate-instances --dry-run` and `Remove-AzResourceGroup -WhatIf` are all judged exactly like the command they simulate, because nothing in `_terraform_destructive/_aws_destructive/the Remove-Az*` matcher inspects those flags. This is a known gap, not a config bug — there is no key that special-cases them. If you need to run one of these unattended, use `report` mode while testing, or approve the one-shot marker the deny message names.
- **An "unpinned" repo makes an ordinary destructive command ask or deny, even when it is obviously fine.** With no `cloud.*` pins, a read-only command (`aws s3 ls`, `az group list`) runs unchecked — but *any* destructive command has an identity the guard cannot vouch for, which is **unknown**, and unknown is never allowed unattended: it **asks** when a person is there to answer and is **denied** when not (`CREW_UNATTENDED=1`, CI, or a non-interactive session). This is the rule most likely to surprise a first-time user, because it fires even in a repo that has never configured anything cloud-related. **Fix — pin the identity this repo may act as**, in `.crew/config.json` (repo-only; a machine-global `cloud` block is ignored):

```
{ "cloud": {  
  
  "awsProfiles": ["acme-dev", "acme-sandbox-*"],  
  
  "awsRegions": ["eu-west-1"],  
  
  "azureSubscriptions": ["00000000-0000-0000-0000-000000000000", "Acme Dev"] } }
```

A profile/region/subscription outside the pins is denied outright (`[cloudIdentity]`). Static `AWS_ACCESS_KEY_ID` credentials and `Remove-Az*`'s saved PowerShell context are **always** unknown — the guard has no way to read either — so pinning does not silence those; only removing the static keys or naming the account through `--profile/--subscription` does.

See [Memory and Obsidian](#) for setup. What goes wrong day to day:

- **Symptom: nothing is being captured / gardened. Check:** is the *primary* vault reachable? Capture, import and the gardener write to the primary vault only, and **never** fall back to a recall vault. If the primary is unavailable — an unmounted drive, a renamed folder — they write nothing and say why on stderr; the session itself is never blocked. `$V0 adopt` shows which vault is currently primary.
- **Symptom: recall never surfaces anything you know is in the vault. Check:**

```
python3 "<crew plugin root>/hooks/scripts/crew_context.py" --stats
```

Read the `miss reasons` line. `cli-missing` means `obsidian-vault` is not installed where `crew` can find it; `cli-exit-N`, `cli-timeout`, `cli-bad-json` mean the CLI ran and failed; `no-vaults` means no vault is currently primary or recall; `no-hits` means the search itself found nothing. Only `no-hits` means "the vault genuinely has nothing relevant" — every other reason is a setup problem. Also confirm `memory.inject` is not false in `.crew/config.json` — recall and code-map injection are on by default since 1.0.0, so "no recall ever appears" with the key set to false is expected. **Fix:** `$V0 recall --query "<words you expect>"` directly, to separate "the CLI can't find it" from "the hook isn't calling the CLI".

- **Symptom: the gardener seems to be falling behind.** It runs in bounded passes on purpose, not continuously: **at most 5 items or 10 minutes per run**, whichever comes first, on **one designated host**, one run at a time. A session whose transcript lives on another machine is left queued for that machine and does not spend one of the five slots. **Check:** `$V0 queue` (what is waiting) and `~/.claude/obsidian/gardener.log` for lines starting `left queued`. **Fix:** run a bounded pass by hand (`/obsidian-vault:garden` or `$V0 garden-run`), or drain a large backlog in batches: `$V0 drain --apply --batches 4`.
- **Symptom: two machines sharing one vault seem to be racing each other.** They are not writing the same file: each host appends to its own `inbox/pending-reflect.<host>.md` and acknowledges to its own `inbox/reflected.<host>.md`, so two machines syncing one vault never contend for the same queue file. An older `inbox/pending-reflect.md` (no host suffix) is still read if present, so an old backlog drains rather than being stranded — it is just never written again.
- **Symptom: `/crew:brainstorm` or `/crew:fix` stops because create refused a ticket id.** On the Obsidian ticket board, `crew_tracker.py create` refuses an id it cannot prove is yours, and prints one of four lines (paths and ids here are examples). The first three begin `id taken`, and `brainstorm` and `fix` then take the

next free id on their own. For any other refusal they stop and show you the line.

- **Another repo owns it.** obsidian: could not update: id taken: T-0060 on Board.md belongs to another repo (<https://example.invalid/team/app>, per T-0060.md), not this one (<https://example.invalid/other/app>): give this repo its own obsidian.boardDir **Fix:** two repos share one board. Give each repo its own obsidian.boardDir.
- **Nobody can tell who owns it.** obsidian: could not update: id taken: could not tell whose card T-0042 is (no T-0042.md note names its repo-id); if it is this repo's, put 'repo-id: <https://example.invalid/other/app>' in T-0042.md **Fix:** if the card is yours, add the line it names to the note (see "The ticket board" in [Memory and Obsidian](#)). If it is not yours, let brainstorm take the next id.
- **Another session in this repo holds it.** files: could not update: id taken: T-0050 is already ready "Fix tests" **Fix:** none needed; .work/INDEX.md already has that ticket. This check runs before the vault is opened, so it wins even when the vault is missing.
- **Crew cannot tell which repo this is.** obsidian: could not update: could not tell this repo's identity: git gave neither an origin URL nor a common dir for <repo path> **Fix:** run crew inside a git checkout. `git rev-parse --git-common-dir` must answer there.

Auto Wrap-Up, Clear and Resume

See [Auto wrap-up, clear and resume](#) for the full cycle that lets a long session finish itself cleanly, clear, and pick up where it stopped — what wrap-up, auto-clear and resume each do, the machine-global keys that turn auto-clear on, and every reason a clear can refuse (logged to `.crew/.autoclear.log`).

"Auto-Clear Did Nothing on Windows"

Symptom: auto-clear is on, a handoff is written and verified, and nothing types the configured command — no keystrokes appear at all.

- **Check context.** `autoClear.method` with `/crew:config --explain` or `crew_config.py --explain`. On native Windows with no tmux pane, "auto" (the default) resolves to "notify", not to any keystroke method — **this is not a bug**, it is the 1.0 owner decision: auto never arms SendKeys on your behalf. "notify" types nothing at all; instead the Stop hook prints a `systemMessage` telling you the handoff is written and verified and it is safe to run the configured command yourself.
- **If you want keystrokes typed for you**, ask for `sendkeys` by name — it is opt-in

only, and `/crew:init`, `/crew:migrate` and `/crew:onboard` all refuse to write it without an explicit yes. It declines and falls back to `notify` (logged to `.crew/.autoclear.log`) inside Windows Terminal specifically, because that host puts every tab in one window and nothing outside it can confirm which tab is active — this is also not a bug, and switching terminal hosts is the only fix.

- **Fix:** either treat the `notify` message as the intended behaviour, or run `crew_autoclear_setup.py apply-method sendkeys --yes` (`${CLAUDE_PLUGIN_ROOT}/hooks/scripts/`) to `opt in`, after reading what `SendKeys` does. See `plugin/crew/CONFIG.md` §14.

"Claude Code Compacted by Itself"

Symptom: the session cleared or summarised itself with no `/clear` or `/compact` typed, and no message about auto-clear ran first.

That is Claude Code's own **built-in auto-compact**, not this plugin. Crew neither causes nor tunes it — `context.autoClear` and `context-watch.sh` only ever act *after* a handoff is written and verified, and nothing in this plugin can trigger a compaction on its own. There is no crew setting that changes when Claude Code's auto-compact fires.

The one thing crew does around either kind of reset: on the next `SessionStart` after any `/clear` or `/compact` — self-initiated or Claude Code's own auto-compact — `handoff-read.sh` reloads `.work/HANDOFF.md` back into context automatically, so a compaction you did not ask for still resumes from the last written handoff rather than from nothing.

Turning Things Off

Every switch named above, in one place. "Off" for a guard means the `PreToolUse` hook still fires but returns immediately without judging anything; "off" for `verifyGate` means the `Stop` hook does nothing at all that turn.

Key	Lives in	Values	What "off"/floor means
<code>memory.inject</code>	repo only	bool	false: no code-map, handoff or vault text is injected (default true since 1.0.0)
<code>guards.cloudGuard</code>	both layers, ratchets	block/report/off	off: the hook reads this key and exits

Key	Lives in	Values	What "off"/floor means
<code>guards.roleWrites</code>	both layers, ratchets	block/ report/off	off: every Write/Edit is allowed unconditionally
<code>guards.terraformApply, forcePush, adminMerge, mergeGate, cloudDestructive, sqlDestructive</code>	both layers, ratchets	block/ask/ allow	there is no "off" — allow is the most permissive tier, still logged
<code>guards.prodDatabase, guards.prodServer</code>	both layers, ratchets	none/read/ full	none is both the default and the floor
<code>scope.mode</code>	repo only	off/report/ block/auto	off: neither the edit guard nor the completion audit runs
<code>scope.allowCliApproval</code>	repo only	bool	false: only a /crew:approve typed by the user counts (no cli or autopilot receipt)
<code>verifyGate</code>	repo only	bool	false: the Stop verify gate does not run at all
<code>context.enabled</code>	both layers	bool	false: context-watch.sh (the Stop nag) does nothing
<code>context.autoClear.enabled, autoWrapUp</code>	both layers	bool	each independently disables one leg of the wrap-up/clear loop
<code>notify.provider</code>	both layers	provider name / none	none: the Notification hook sends nothing
<code>change.requireForProduction</code>	both layers, ratchets (may only turn ON)	bool	false: promoting needs no approved change request

Key	Lives in	Values	What "off"/floor means
install.policy	both layers, ratchets	see plugin/crew /CONFIG.md \$"install"	narrowest tier refuses more install actions

~/.claude/crew/config.json is the machine-global layer; .crew/config.json is per-repo and wins where both speak. A **ratcheted** key (marked above) can only be *narrowed* by the repo relative to the machine-global value, never widened — a repo cloned from someone else cannot silently loosen a guard the machine owner set to block. /crew:config --show shows where each setting actually came from, and /crew:config with no argument (alias /crew:config-setup) sets either layer from a menu: it marks a repo value that widens a guard with !, and names a repo value the machine layer holds down. The same menu can delete a repo's .crew/config.json: it moves the file to a backup in one rename, compares it with what the preview read (a file that changed since is put back and nothing is deleted), and prints the command that restores it for sh, cmd and PowerShell.

Web Testing

Symptom: --select web-testing (or the ticked default row) fails, or Playwright runs but the visual rule never passes.

- **Node too old.** install-prerequisites.{sh,ps1} checks for Node >= 20.19 (or = 22.12) itself and refuses rather than installing or upgrading Node for you — node -v, install a newer one, and re-run --select web-testing.
- **Browsers missing system dependencies.** npx playwright install --with-deps chromium is what the installer runs; a bare npx playwright install (no --with-deps) skips the OS packages Chromium needs and produces a browser that launches, then crashes on first navigation.
- **Docker absent -> visual UNVERIFIED, not a fail.** toHaveScreenshot baselines are only meaningful when generated inside mcr.microsoft.com/playwright:v1.63.0-noble — the installer warns rather than blocking when Docker is missing, and the visual verify.json rule is written to skip (report UNVERIFIED) outside that image rather than compare against a baseline that can never match.
- **MCP server won't connect on Windows.** Both @playwright/mcp and chrome-

devtools-mcp are launched with npx, which the MCP host cannot invoke directly on Windows — wrap the entry as "command": "cmd", "args": ["/c", "npx", ...] in .mcp.json (documented in chrome-devtools-mcp's own troubleshooting for the "Connection closed" failure); the stack-web skill's .mcp.json example already shows this.

Windows Notes

- **pwsh is not on Git Bash's PATH.** Every script that shells out to PowerShell names it by an absolute path or resolves it explicitly; a bare pwsh in a hand-typed command fails as "command not found", which is easy to misread as PowerShell itself being broken.
- **Git Bash ships without python3.** Every hook wrapper here resolves python3, then python, then py, and fails loudly on stderr rather than silently standing down when none resolves (see "no-python fail-closed", above). If a hook you expect to run says nothing at all, check for a usable Python on PATH first.
- **Both flavours are always registered.** hooks.json pairs every bash command with a shell: "powershell" sibling on the same event, so a machine with both bash and pwsh present runs both. This is not double-firing to fix — it is how the same hook reaches both shells; each PowerShell twin stands itself down on the wrong platform (\$env:OS -ne 'Windows_NT') rather than the bash twin doing that job.
- **Tests and checks are slow on Windows.** Read /crew:status's shell line first. It names the shell route crew's jobs take (shellRoute.mode: auto, wsl, powershell, gitbash) and why. WSL never probed means nobody ran crew_shell.py probe --write; not-installed comes with the wsl --install -d Ubuntu recommendation, which crew prints and never runs. A repo on a Windows drive goes to WSL under auto only after crew_shell.py measure --write has shown WSL faster there. See "Choosing the shell route on Windows" in plugin/crew/skills/crew-setup/platform.md.

Auto Wrap-Up, Auto-Clear and Auto-Resume

This section is folded into the troubleshooting guide (troubleshooting.md), linked from it as "Auto wrap-up, clear and resume". It covers the cycle that lets a long session finish itself cleanly, clear, and pick up where it stopped.

What the Cycle Does

1. **Wrap-up.** When the context window crosses its threshold, the Stop hook blocks

once. It asks the session to finish or park the change in flight, write `.work/HANDOFF.md`, and update the ticket. This happens once for each threshold crossing in each session. It never happens on the forced continuation that follows a block. With `context.autoClear.wrapUp` armed (below), the block carries the wrap-up procedure instead, and the clear waits for its results.

2. **Clear, or notify.** After the handoff is written, auto-clear acts on it. What it does depends on method (below): `notify` (the default on native Windows) prints a message telling you the handoff is written and it is safe to run `/clear` (or `/compact`) yourself — it types nothing. `tmux`, `xdotool` and `sendkeys` actually type the configured command into the terminal that runs this session. Whichever method is in force, it acts only when all five of these hold: - this machine opted in; - the handoff was written by this session after the wrap-up request, is longer than a stub, and is not the automatic PreCompact skeleton; - the context reading that caused the wrap-up was a real measurement (see "When it will not clear"); - this session is bound to its own Claude Code process, and that process has a terminal of its own (see "Which window gets the keystroke") — a headless session, such as a `claude -p child`, gets one message instead, naming its handoff and saying the process that started it must start a new one; - for a method that types (`tmux/xdotool/sendkeys`), the target window can be identified exactly, from that process — `notify` needs none of this, since it identifies no window.

Crew neither causes nor tunes Claude Code's own auto-compact. That is a separate, built-in behaviour that fires whenever Claude Code itself decides to; this cycle only reacts to a handoff once one exists, and nothing here makes auto-compact happen sooner, later, or at all. 3. **Resume.** The new session starts with the handoff and its next action already in context, within the 3,000-character resume budget. Nothing starts working unprompted: crew does not use `initialUserMessage`, so you press Enter to continue. Claude Code 2.1.282 drops it in an interactive session anyway (spike, 2026-09-25): it is honoured by `claude -p` only.

With `resume.auto: true` in the machine-global config, the handoff after `/clear` or a manual `/compact` also names the exact next command, read from the note's `resume:` line (`resume: /crew:done T-0001`): Auto-resume: ready to run `/crew:done T-0001`. — or Auto-resume did not start: `<reason>`. when a check refuses it. It does not start from the hook: you press Enter or type it, or crew types it where the terminal can be identified (below). The refusal reasons are: compact was not a manual `/compact`; no handoff note, or it was archived as stale, or is stale and could not be archived; the handoff is the automatic PreCompact skeleton; no resume line, `resume: none`, or a line the grammar refuses (a second line, trailing text, an unknown command, or an excluded one such as `/crew:approve`); the note's `branch:` or `head:` does not match the checkout; the ticket directory or goal file is missing; the command is not installed; `handoff-author.json` could not be read; no record of which session wrote this handoff; the handoff changed since its author session wrote it; the handoff was written by another session; this session's

process could not be identified (always on a host without /proc, such as native Windows or macOS); the record of past auto-resumes (resume-state.json) could not be read, or its directory cannot be searched; this handoff was already resumed; the progress fingerprint could not be computed; the same command with no progress since the last auto-resume; internal error (the decision itself failed, and the handoff is still injected). Never on a plain startup.

Crew types it where it can (T-0013). The context hook runs its own flavour's auto-clear sender in resume mode. Consent is resume.auto; context.autoClear only describes the terminal. In tmux it waits resume.typeDelaySeconds (default 2), polls the pane for an idle, empty input line for up to resume.readyTimeoutSeconds (default 15), then sends the command and, half a second later, Enter. On native Windows it types only under an explicit method: sendkeys, after the delay, with the focus and tab rechecks; auto there types nothing. It types once per handoff (a marker under <git-common-dir>/crew/) and records the run first, so a run that could not be recorded is never typed. The context then says Auto-resume: typing /crew:done T-0001 into this session in 2s (method tmux), or keeps the line above and adds Auto-resume was not typed: <reason>. Every refusal is in .crew/.autoclear.log: decide did not say run; no usable method, \$TMUX unset or a pane that is not this session's; wtype or xdotool; already typed; the run could not be recorded; the input line is not empty; not ready in time; on Windows, focus lost or a tab not provable. The 2 s default comes from a measurement (input ready by 0.134 s after SessionStart on Claude Code 2.1.282); a delay is still a guess, and Windows has no probe.

A note resumes only in the session that wrote it. Write it with the Write tool (as /crew:handoff does): the context hook then records which session and which Claude Code process wrote it. After /compact the session must match; after /clear the process must, so a /clear in another terminal on the same worktree waits. A note written by Bash or by hand waits.

Turning It on and Off

Wrap-up and resume are on by default. Auto-clear is **off** by default. You turn it on per machine, in the machine-global config, because it drives this machine's keyboard:

Where	Key	Value	Effect
~/ .claude/ crew/ config.json (Windows: %USERPROFILE% %\ .claude\crew\ config.json)	context.autoClear.enabled	true	turns auto-clear on for every crew repo on this machine

Where	Key	Value	Effect
same	<code>context.autoClear.method</code>	"auto" (default), "tmux", "xdotool", "notify", "sendkeys" , "none"	how the cycle acts — see "Which method, and what it does" below
same	<code>context.autoClear.windowTitle</code>	a substring of the terminal's title	fallback when the window cannot be found by process id
same	<code>context.autoClear.delaySeconds</code>	3	wait before typing, so the prompt exists
same	<code>context.autoClear.minHandoffLines</code>	5	a shorter handoff counts as a stub
<code>.crew/ config.json</code> (the repo)	<code>context.autoClear.enabled</code>	false	switches auto-clear off in this repo. A repo can switch it off, but true here does not switch it on.
<code>~/ .claude/ crew/ config.json</code>	<code>resume.auto</code>	true	names the next command from the handoff's resume: line after <code>/clear</code> or a manual <code>/compact</code> (off by default)
<code>.crew/crew.json</code> or <code>.crew/config.js</code> on (the repo)	<code>resume.auto</code>	false	vetoed auto-resume in this repo. true here does not switch it on.
<code>~/ .claude/ crew/ config.json</code>	<code>context.autoClear.wrapUp</code>	true	turns the wrap-up into the wrap-up

Where	Key	Value	Effect
			procedure and makes the clear wait for its results (off by default; acts only where enabled is armed). A repo false vetoes it; a repo true does nothing.
.crew/ config.json	context.autoWrapUp	false	replaces the wrap-up instruction with a plain handoff request. It still blocks once. Armed wrapUp supersedes both wordings.
.crew/ config.json	context.enabled	false	turns off the whole context watcher, wrap-up included
.crew/ config.json	memory.inject	false	handoff-read prints the handoff for you to read, without extracting the next action. context.autoResume is no longer read.

/crew:config --show shows where each value comes from. /crew:config with no argument (or /crew:config-setup) opens a menu that sets either file from a list of values, with a dry run before anything is written. To turn the cycle off on a machine, set context.autoClear.enabled to false or delete the key. In a repo the menu offers only the veto (false) or null for context.autoClear.enabled and resume.auto, and shows onlyRepos/onlySessions read-only, because only the machine file can arm or narrow them.

Armed wrap-up (context.autoClear.wrapUp)

A hook cannot make the session do anything; it can only refuse to clear. So with wrapUp armed the threshold block sends one procedure — start no new step; run the step's Test: command (the active ticket's plan step, or the checks for the tracked diff with no ticket);

commit only if it passes; if it cannot pass, leave the tree and write `resume: none` with the reason under **Verify first**; run `/crew:handoff --wrap-up`; end the turn — and auto-clear acts only when, beside the checks above:

- the handoff's head: is HEAD, so it was written after the commit;
- its branch: is the checked-out branch;
- no tracked file is modified (untracked files, and the handoff itself, do not count);
- its resume: line parses, or is `resume: none`.

Anything crew cannot tell — git failing, no `crew_resume`, on native Windows no python — refuses. A refusal is logged, shown to you as `crew wrap-up: not clearing - <reason>`, and handed back to the session once, at its next ordinary Stop. Crew checks that the commit happened, not that the test passed: the verify gate stands down on the turn the commit is made on.

The chain end to end: wrap-up (T-0017) → the target proven from this session's own process (T-0016) → `clear` → `resume.auto` decides (T-0006) → the command is typed (T-0013).

Arming one scratch repo while other sessions are live

`enabled: true` arms auto-clear in **every** Claude session on the machine, not just the one you are testing. If other sessions are running, one of them can get `/clear` typed into its terminal when its own handoff lands. To limit auto-clear to one repo or one session, add these keys to the same machine file. They can only narrow what `enabled` turns on:

Key	Value	Effect
<code>context.autoClear.onlyRepos</code>	list of absolute repo paths	auto-clear runs only in these repos
<code>context.autoClear.onlySessions</code>	list of session ids	auto-clear runs only in these sessions

- A missing key or `null` does not narrow anything. An empty list `[]` turns auto-clear off everywhere. If a key holds anything other than a list, auto-clear is off everywhere too.
- When both keys are set, a session must match both.
- Paths are compared after symlinks are resolved. Backslashes and trailing separators do not matter. On Windows, case does not matter and `/c/repos/x` means `C:\repos\x`. A relative path such as `.` never matches.
- Session ids must match exactly, including case.
- These keys are read **only** from the machine file. If a repo's `.crew/config.json` sets

them, they are ignored, so a repo cannot add itself. A repo's `enabled: false` still turns auto-clear off.

- In any other repo or session, auto-clear stays silent. It does not write a log line, just as it does when `enabled` is off.

To arm one scratch repo safely:

```
{
  "context": {
    "autoClear": {
      "enabled": true,
      "onlyRepos": ["C:\\\\repos\\\\autoclear-scratch"]
    }
  }
}
```

On Linux, use a path such as `/home/you/autoclear-scratch`. When you are done testing, remove `enabled` or set it to `false`. Also delete `onlyRepos`, so that a later `enabled: true` does not arm only that repo without anyone noticing.

Which Method, and What It Does

Method	What it does	Where auto picks it
tmux	Types the command into the <code>\$TMUX_PANE</code> whose process is this session's own process or an ancestor of it. No focus involved — the most reliable method.	Linux, macOS, WSL, Git Bash on Windows, when <code>\$TMUX</code> names a pane and <code>tmux</code> is on <code>PATH</code>
xdotool	Types into the one X11 window owned by the nearest ancestor of this session's own process, when that window hosts no other terminal (or matching <code>windowTitle</code>).	Linux with <code>xdotool</code> on <code>PATH</code> and <code>\$DISPLAY</code> set

Method

What it does

Where auto picks it

notify

Types nothing. Prints a message saying the handoff is written and verified and it is safe to run the command yourself. Never says anything was cleared or compacted — it does not know that, and it did not do it.

native Windows, whenever there is no tmux pane — this is the default there, not a fallback for something missing

sendkeys

`System.Windows.Forms.SendKeys` against the one window owned by an ancestor of this session's own process (or matching `windowTitle`), confirmed still in the foreground right before typing. **auto never picks this** — typing into a window this hook found itself is a risk auto does not get to accept on your behalf. Set method: "sendkeys" explicitly to opt in. Even then, it declines and falls back to `notify` — logging why — when the target window is owned by Windows Terminal, because that host puts every tab in one window and nothing outside it can confirm which tab is active.

never (opt-in only)

none

Refused outright.

never

When It Will Not Clear (or Notify)

Each refusal is logged with its reason to `.crew/.autoclear.log`. A Stop hook's stderr is not shown to you, so check that log first. A `notify` or `sendkeys` decline is logged the same way — "declined" rather than "refusing", since the machine has opted in and something still happened (a message, or a fallback), just not a keystroke.

Log says

Meaning

not trustworthy (estimated-from-transcript-size)

no usage record was found, so the reading was an estimate

not trustworthy (unknown-window)

the model's context window is unknown and `context.budgetTokens` is not set

not trustworthy (stale-reading-before-

the last reading came from before a

Log says	Meaning
compaction)	compaction
the wrap-up marker is empty or not JSON	the marker was not written by this version of the hook. Nothing clears on it.
predates this session's wrap-up request	the handoff on disk is older than the request, probably left by an earlier session
automatic PreCompact skeleton	only the compaction skeleton exists, not a real handoff
N windows have a title containing ... / has N windows	the window is ambiguous, so nothing is typed (sendkeys/xdotool only)
no window belongs to any ancestor of this hook / ... of this session's process	the terminal's window could not be found by process id, and no windowTitle is set (sendkeys/xdotool only)
sent - method notify-headless: crew: this session has no terminal of its own (...)	a headless session (claude -p: entrypoint sdk-cli, or no controlling terminal): nothing was typed, and the message names the handoff and its resume: line. The process that started the session restarts it
could not identify this session's process (...)	no Claude Code session record names a process above the hook with this session's id and start time (or the record is unreadable, or names another session). Nothing is typed; auto falls back to notify
could not tell whether this session has a terminal of its own (entrypoint '...' ...)	the session record's entrypoint is not one measured to have a terminal (only cli is). Nothing is typed; auto falls back to notify
passes through another Claude Code session (pid N)	this session is a child of another session, and the pane or window above it is the parent's
the window's owner (pid N) also hosts another terminal / another Claude Code session	the terminal that owns the window also runs another tab or session, so the window is not provably this one's
could not be listed / the parent of pid	the process tree could not be read far

Log says

Meaning

N could not be read / deeper than 16 processes

enough to prove anything

found by its title alone, and another Claude Code session is live / has no owning process

a windowTitle match (or a window with no process) cannot tell two sessions apart

tmux pane %N could not be confirmed

\$TMUX_PANE is not the pane that runs this session

method wtype ... is refused

Wayland's wtype cannot target a window. Use tmux instead.

sent - method notify

the message was printed — this is the normal, successful outcome for notify

declined sendkeys - cannot verify the active tab ...

the target window is owned by Windows Terminal; sendkeys fell back to notify instead of guessing which tab is active

declined sendkeys - the target window lost focus during the ...s delay

the window had focus when identified, but not when the delay ended, so nothing was typed

Two sessions in one repository no longer interfere with each other. Wrap-up markers are .crew/.handoff-requested-<session_id>, and a SessionStart clears only its own session's marker.

Which Window Gets the Keystroke

Applies only to the methods that type — tmux, xdotool, sendkeys. notify identifies no window and needs none of this.

First, the session is bound to its own process: the nearest ancestor of the hook named by a Claude Code session record (\${CLAUDE_CONFIG_DIR:~/.claude}/sessions/<pid>.json) carrying this session's id and that process's start time. Only a session whose record says kind interactive and entrypoint cli, with a controlling terminal (native Windows has none to read), is typed into. Every rule below walks up from that process, not from the hook, and refuses when the walk passes through another Claude Code session — a child started from a session's Bash tool inherits \$TMUX, and without this its parent's pane is the one that got /clear. CONFIG.md §14 has the full rules and their limits (Windows' entrypoint is unmeasured; macOS cannot check the start time).

- **tmux (Linux, macOS, WSL):** the pane in `$TMUX_PANE`, and only when that pane's process is this session's own process or an ancestor of it. No title is needed. This is the most reliable method.
- **X11 (xdotool):** the one visible window owned by the nearest ancestor of the session's process, and only when no other terminal (another tab's tty, another session) runs under that window's owner. If that process owns more than one window, `windowTitle` has to narrow the choice to one. If no ancestor owns a window, exactly one window must match `windowTitle`, and no other session may be live, since a title cannot tell two apart. The keystroke goes to that window id after it is activated and confirmed active.
- **Windows (sendkeys):** chosen the same way, by owner process first and then by title. Declines outright, before any of that matters, when the resolved window is owned by Windows Terminal — see the method table above. Otherwise, after the delay, the command is sent only when that exact window handle still has focus; if it does not, nothing is typed and the decline is logged.

Windows Terminal runs every tab in one window. A process-id match cannot tell which tab is active, so on Windows Terminal set `windowTitle` to something only this session's tab shows.